

Async Landmines in the AI Stack: An AST-Based Empirical Study of Concurrency and Resource-Management Defects in Popular Python AI/ML Frameworks

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Abstract—Modern AI and machine-learning systems are overwhelmingly implemented in Python and depend heavily on the `asyncio` event loop for high-throughput model serving, agent orchestration, and tool execution. Asynchronous Python, however, is susceptible to a family of subtle, hard-to-debug defects—synchronous calls that block the event loop, fire-and-forget tasks that may be garbage-collected before completion, leaked file descriptors, mutable default arguments, and deprecated event-loop APIs—that routinely pass unit tests and code review while degrading reliability in production. We present `codehound`, a zero-dependency, AST-based static analyzer (≈ 750 LOC, standard-library ast only) that detects six such defect classes, and we apply it in a systematic empirical study across 29 widely-used open-source Python AI/ML frameworks (combined >1.28 million GitHub stars, 6.4 million lines of Python). We identify 1,377 defect instances in 24 of the 29 frameworks, dominated by mutable default arguments (53.9%), deprecated event-loop usage (20.8%), and fire-and-forget tasks (19.0%), with rarer but higher-severity event-loop-blocking calls clustered in HTTP serving routes and agent loops. Crucially, we validate a sample of detections by submitting upstream fixes: five pull requests were merged by maintainers (into `vLLM`-, `HuggingFace`-, `agno`-, and `mem0`-class projects), providing real-world ground truth that these are genuine, accepted defects rather than stylistic preferences—and we analyze the false positives (e.g., `get_event_loop()` invoked inside an already-running loop) that a precise detector must avoid. Our results indicate that `async`-safety and resource-management anti-patterns are prevalent even in mature, heavily-used AI infrastructure, and that lightweight AST analysis surfaces them at high practical precision.

Index Terms—static analysis, `asyncio`, empirical software engineering, machine-learning infrastructure, software reliability, abstract syntax trees

I. Introduction

The infrastructure that the contemporary AI industry runs on—model-serving engines, agent frameworks, retrieval pipelines, and LLM tool integrations—is written predominantly in Python and increasingly built on `asyncio` to achieve the concurrency that high-throughput inference and multi-agent systems demand. Yet asynchronous programming concentrates a class of defects that are

notoriously difficult to detect dynamically: they do not raise exceptions, they pass tests, and they often manifest only under production load or specific garbage-collection timing.

Consider three representative examples observed in widely-deployed projects. (i) A model-export route declared `async def` repeatedly invokes `time.sleep(0.5)` in a polling loop—blocking the entire event loop for up to 30 seconds while every other in-flight request stalls. (ii) A request-abort endpoint schedules the abort with a bare `asyncio.create_task(...)` and discards the returned task; because the loop retains only a weak reference, the task may be garbage-collected before the abort executes, so the abort silently never happens. (iii) A library function defaults a parameter to a mutable `{}`, sharing one dictionary across all invocations. None of these are caught by tests; all are real reliability hazards.

This paper asks how prevalent such defects are across the popular Python AI/ML ecosystem, and whether lightweight static analysis can detect them with sufficient precision to be useful. We make three contributions: (1) `codehound`, an open-source, zero-dependency AST analyzer detecting six concurrency- and resource-related defect classes; (2) a systematic prevalence study across 29 popular frameworks; and (3) a real-world validation in which we submit detections as upstream pull requests and use maintainer acceptance (merge) as ground truth.

II. Background and Related Work

The `asyncio` execution model. `asyncio` is cooperative single-threaded concurrency: one event loop interleaves many tasks, switching only at `await` points. Between `awaits`, whatever runs monopolizes the loop. A synchronous blocking call freezes all concurrency; a task the loop does not strongly reference may be reclaimed mid-execution [1].

Defect classes. We study six patterns: blocking calls in `async` (CH001), mutable default argu-

ments (CH002, cf. flake8-bugbear B006 [3]), deprecated `datetime.utcnow()` (CH003), deprecated `asyncio.get_event_loop()` (CH004), unclosed file handles (CH005), and discarded `create_task/ensure_future` results (CH006, cf. Ruff RUF006 [2]).

Related work. General-purpose linters (Ruff, flake8-bugbear, pylint) and type checkers (mypy) detect subsets of these patterns. Prior empirical work has studied concurrency bugs in Python and ML-system reliability broadly. To our knowledge, no prior study (a) focuses specifically on async-safety and resource defects in the AI/ML framework ecosystem, (b) measures prevalence at scale with a reproducible lightweight analyzer, and (c) validates findings against real maintainer acceptance of fixes.

III. The codehound Analyzer

codehound parses each source file with the standard-library `ast` module and precomputes a child→parent map over the tree (Python’s `ast` does not record parents). This map enables cheap upward scope queries that the checks rely on: what function encloses this node?, is this call the operand of an `await`?, and is this statement inside a `with` block?. Each defect class is an independent Check returning `Finding(path, line, col, code, message)`. The analyzer is ≈ 750 LOC, has zero third-party dependencies, and is released open-source.

Two precision choices are notable. CH001 ignores awaited calls (`await client.get(...)` does not block, even if the receiver name resembles a synchronous library). CH006 flags only bare expression-statement calls to task creators (result not assigned, awaited, or returned), avoiding tasks that are stored or handed to a `TaskGroup`.

IV. Study Design

Corpus. We select 29 popular open-source Python AI/ML frameworks (model serving, agent frameworks, LLM tooling, RAG, training) from GitHub. Each repository is shallow-cloned and pinned to a specific commit. We exclude test, example, and documentation directories, focusing on production code.

Scan protocol. For each repository we run all six checks and record per-class counts and Python LOC.

Validation protocol. Static counts alone cannot establish that a flagged pattern is a bug worth fixing. We submit a sample of detections as upstream pull requests and record outcomes—merged (maintainer-confirmed true positive), open/under-review, or rejected—yielding a real-world precision signal and a false-positive taxonomy.

V. Results

RQ1 (Prevalence). Across 29 frameworks (>1.28 M combined stars, 6.4M LOC) codehound flags 1,377 defect instances, present in 24 of the 29 frameworks; only five returned clean. Aggregate defect density is 0.22 per KLOC. The anti-patterns are systemic—present even in

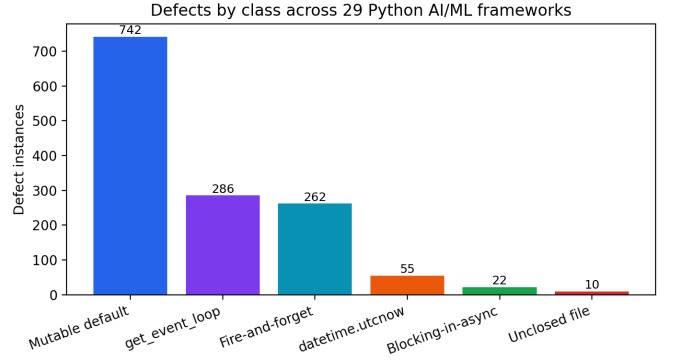


Fig. 1. Defect instances by class across the 29-framework corpus.

TABLE I
Defects by class (full corpus).

Code	Defect	Count	Share
CH002	Mutable default argument	742	53.9%
CH004	Deprecated <code>get_event_loop()</code>	286	20.8%
CH006	Fire-and-forget task	262	19.0%
CH003	Deprecated <code>datetime.utcnow()</code>	55	4.0%
CH001	Blocking call in async	22	1.6%
CH005	Unclosed file handle	10	0.7%
Total		1,377	100%

projects from major industrial labs (Microsoft `autogen`: 95; HuggingFace `transformers`: 45).

RQ2 (Distribution). Table I and Fig. 1 show the breakdown. Mutable default arguments dominate (53.9%); the concurrency-specific classes (CH004 + CH006) account for 39.8%; event-loop-blocking calls (CH001) are rare (1.6%) but highest-severity and cluster in HTTP serving routes and retry loops. One framework (litellm) is a pronounced outlier (536 defects, 39% of the total).

RQ3 (Validation). We submitted a sample of detections upstream (Table II). Five fixes were merged by maintainers, confirming the flagged patterns as genuine, accepted defects in heavily-used projects; eleven remain under review.

RQ4 (False positives). Honest precision analysis requires the rejected/not-submitted cases. (i) `get_event_loop()` inside a running loop (dspy #9907, rejected): valid there and emits no warning—a genuine false positive motivating context-sensitivity for CH004. (ii) Intentionally-suppressed blocking calls (langflow, # noqa: ASYNC251): a knowingly-accepted-risk category. (iii) Non-mutated mutable defaults (chroma): latent rather than active. (iv) Deliberate fire-and-forget utilities (camel): by design. Raw counts thus mix active bugs, latent footguns, and accepted risks; the merged PRs (RQ3) anchor the active-bug end.

VI. Discussion

That defects appear in 24 of 29 analyzed frameworks—including those maintained by major industrial labs—

TABLE II
Validation outcomes (real upstream pull requests).

Repo	Class	PR	Outcome
unsloth	CH001	#6135	Merged
agno	CH001	#8158	Merged
agno	CH001	#8186	Merged
agno	CH005	#8161	Merged
mem0	CH002	#5302	Merged
vLLM	CH006	#45249	Under review
autogen	CH006	#7825	Under review
sglang	CH001	#28029	Under review
jina	CH001	#6243	Under review
khoj	CH001	#1342	Under review
OpenAI Agents	CH006	#3553	Under review
litellm	CH006	#29417	Under review
ragas	CH004	#2757	Under review
agno	CH006/4	#8183	Under review
Future AGI	CH006	#821	Under review

suggests these hazards are a tooling gap rather than a knowledge gap: standard CI does not reliably surface them, and they survive review because they are invisible at rest. The clustering of the highest-severity class (CH001) in serving routes and retry loops has direct implications for ML-serving reliability: a single blocking call on a hot path can collapse the concurrency of an entire inference server. The five clean projects show the patterns are avoidable with appropriate linting discipline. We argue AST-level checks of this kind belong in the default CI of AI/ML libraries.

VII. Threats to Validity

Internal. Static analysis incurs false positives/negatives; we mitigate via the precision analysis (RQ4) and anchor true-positiveness with maintainer merges (RQ3). The submitted PRs are a non-random sample, so the merge rate is an existence proof of true positives, not an unbiased precision estimate. External. The corpus, though popular, may not generalize to all Python ML code. Construct. Maintainer acceptance proxies “real bug worth fixing,” but non-merge can reflect process (e.g., required pre-approved issues) rather than invalidity.

VIII. Conclusion

Async-safety and resource-management anti-patterns are pervasive across popular Python AI/ML frameworks, and a lightweight, zero-dependency AST analyzer surfaces them with demonstrated real-world utility—five fixes merged into widely-used projects. Future work: extend the corpus and checks, add context-sensitivity to reduce false positives (CH004), measure per-KLOC density vs. project age/size, and integrate codehound into CI to study defect lifecycles.

References

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